

Eduardo Gaitán Escalante

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EDUCATION

Universidad Nacional Autónoma de México | Ciudad Universitaria, CDMX, México

Oct 2024

B.A. in Economics, graduated with honors | GPA: 9.4/10.0

Relevant Coursework: Econometrics, Statistics, Time Series, Mathematics, Linear Algebra, Game Theory, Economic Development.

Certifications: Machine Learning with Apache Spark & PySpark; Databricks Fundamentals; Geographic Data Science with Python (Center for Spatial Data Science - University of Chicago); Intro to Deep Learning & Time Series - Kaggle; Learning Amazon SageMaker - LinkedIn Learning; Unlock Your Data with Data Cloud - Salesforce.

TECHNICAL SKILLS

Programming & Data: Python (advanced), R (advanced), JavaScript/TypeScript, SQL, PostgreSQL, SQLite, Scikit-learn, TensorFlow, GeoPandas, Parquet.

Applications & Integrations: FastAPI, Flask, Next.js, Astro, React, python-telegram-bot, Google Sheets API, Gemini, Amadeus, Stripe, n8n, webhooks.

Infrastructure & Operations: Docker, Docker Compose, Traefik, Linux, GitHub Actions, Hostinger VPS, HTTPS/TLS routing, systemd, PM2, UFW, fail2ban, Trivy.

IT Operations: PC and server hardware/software installation and configuration; endpoint and peripheral support; hardware/software inventory control; cloud service support.

Operating Systems: Windows 10/11, Linux.

RELEVANT EXPERIENCE

Freelance Systems & Automation Engineer

Jan 2025 - Present

- Designed and deployed client-facing digital operations including Telegram lookup, sales and commission bots, a WhatsApp service platform, an e-commerce storefront, flight monitoring, and VPS observability.
- Automated workflows with n8n, Google Sheets, API/webhook integrations, and Telegram notifications; implemented caching, retries, multi-step state machines, health checks, and audit logs.
- Containerized services with Docker/Compose and Traefik, operating HTTPS routes on a Hostinger VPS and documenting repeatable deployment, rollback, and service-verification patterns.
- Built Sentinel monitoring for service health, system resources, certificates, SSH/firewall posture, exposed ports, security updates, and Trivy CVEs, with approval-gated remediation actions.

Naturgy | Data Science Specialist

Jan 2025 - Present

- Led a cross-functional team in designing and deploying an **Isolation Forest** model for anomaly detection, including ETL pipelines, automation workflows, orchestration, hyperparameter tuning, and performance analysis using H2O and R.
- Modernized a legacy **K-Means** customer-segmentation model, achieving a 15% improvement in accuracy and a 10% reduction in training time.
- Partnered with Economics and Finance to automate 7 manual processes, reducing processing time from 5 hours per task to under 1 minute and saving hundreds of hours annually.
- Proposed and developed a hybrid boosted forecasting model (XGBoost + seasonality) for annual natural-gas consumption.

National Council of Evaluation of the Social Development Policy | Head of the Small Area

Jan 2024 - Dec 2025

Estimation Methodologies Department

- Led a data-pipeline project using Docker, Kedro, PostgreSQL, pgAdmin4, and Git, improving storage and analysis workflows.
- Fitted supervised and unsupervised models including Random Forest, XG/Gradient Boosting, Regression/Logistic, SVM, neural networks, and hybrid boosted time-series models.
- Used SQL to manage queries and manipulate datasets efficiently in pgAdmin4; contributed to poverty-analysis publications within the technical estimation team.

RELEVANT EXPERIENCE (CONTINUED)

General Coordination for Planning and Investment - JGCDMX | Environment, Tourism, and Economic Development Coordinator Oct - Nov 2023

- Automated ETL processes with Python scripts, reducing project-creation times by 20% and making information available to land teams sooner.
- Developed interactive geospatial dashboards using GeoPandas, Leaflet, and Folium for decision-making and route planning.
- Deployed two geospatial models (K-means and Spatial Lag Model) for heatmaps, analytical reporting, and pattern distribution.

Python & R Developer | Program for Development Studies - UNAM Jan 2020 - Sep 2023

- Led the creation of a geospatial night-time-light data pipeline, optimizing analysis, storage, and reporting with Parquet.
- Designed Docker containers for developer environments integrating PostgreSQL databases, Python, and R tools.
- Conducted advanced geospatial estimations for socioeconomic research purposes.

SELECTED FREELANCE PROJECTS

Customer & Operations Automation Platform | GitHub: telegram-bot-agency 2025 - 2026

- Built a Telegram client-data lookup service and extended the platform with multi-tenant WhatsApp handlers for product questions, orders, QA, and guided flight search.
- Added Google Sheets product access with caching, asynchronous wrappers, order state machines, Amadeus integration, Docker packaging, Traefik routing, and a tested FastAPI service.

Punto Clave MX E-commerce | GitHub: spike-ecommerce-web 2026

- Developed and deployed a Next.js/TypeScript storefront with product management, Stripe Checkout, SPEI fallback scaffolding, media routes, SEO, and GitHub Actions over SSH.

VPS Sentinel & Operations Hub | Private client infrastructure 2026

- Implemented health and security monitoring for Docker services, systemd units, HTTP/TLS endpoints, disk/memory, SSH hardening, UFW, fail2ban, CrowdSec, and container CVEs.
- Added operator-readable Telegram alerts and a kill-switched approval layer for a small allowlist of deterministic remediation actions.

Payment Reminder System | GitHub: payments-reminder 2025

- Created an automated reminder workflow using RFM segmentation, ML-based channel selection, WhatsApp/Twilio messaging, Notion, and scheduled GitHub Actions execution.

ADDITIONAL PROJECTS

Databricks Experiment | Personal Project Dec 2024

- Built a Gradient Boosting classifier for air-quality labels and configured MLflow for experiment tracking, metrics logging, and model-performance optimization.

Estimation of Poverty Using Nighttime Satellite Light Data | Academic Project Jan 2023

- Implemented SLX, spatial-lag, and spatial-error-lag models to examine the relationship between luminosity and poverty levels, identifying an inverse correlation.